

AI Keyword Search Analysis Project Write-Up

Using Python and SQL, I analyzed Google Search ad performance data for AI-related keywords, focusing on impressions, click-through rate (CTR), and average cost-per-click (Avg CPC).

A bar chart of average CTR by match type showed *Exact Match (Close Variant)* leading, followed by *Broad Match* and *Phrase Match (Close Variant)*. Standard *Phrase Match* and *Exact Match* trailed, suggesting that close variants capture more natural searches and drive better engagement.

An [interactive scatter plot](#) compared CTR (y-axis), Avg CPC (x-axis), and impressions (bubble size). Key patterns emerged:

- **Top-left (High CTR, Low CPC):** Ads here get clicked often without costing much per click. For example, “*AI coding assistant*” (Exact Match) and “*AI for data analysis*” (Broad Match) both attracted attention at a low price, making them strong candidates for scaling.
- **Top-right (High CTR, High CPC):** Ads get strong engagement, but each click is expensive. “*AI tools*” (Phrase Match) shows that users are clearly interested, but the high CPC means the cost could outweigh the benefit unless optimized.
- **Bottom-left (Low CTR, Low CPC):** Ads don’t get many clicks, but they’re cheap to run. “*AI software*” (Broad: 83 impressions, Phrase: 670 impressions) shows that even though it shows up often, very few people click, limiting its value despite the low cost.
- **Bottom-right (Low CTR, High CPC):** Ads here don’t attract clicks and are costly when they do. “*AI apps*” (Exact Close Variant) and “*AI tools*” (Exact Match) fall in this quadrant, making them the least efficient and best suited for pausing or reducing spend.

Overall, the analysis shows that match type and keyword choice strongly influence both cost and engagement. By separating keywords into efficient and inefficient quadrants, it becomes clear where to invest more, where to test, and where to reduce spend. This approach helps ensure advertising budgets are used more effectively and deliver stronger returns.

Specific keyword-level insights highlight these patterns in practice. Exact Match (Close Variant) for terms like “AI coding assistant” performed especially well, combining high CTR with low CPC, making it a strong candidate for scaling. Broad Match also surfaced valuable terms such as “AI for data analysis,” which balanced reach with cost efficiency. In contrast, Phrase Match keywords like “AI tools” showed strong interest but at a high cost per click, while “AI software” underperformed despite generating high impressions. For companies, these findings translate into clear actions: prioritize scaling high-engagement, low-cost keywords, cut back on costly or low-performing ones, and refine match-type strategies to achieve better ROI.